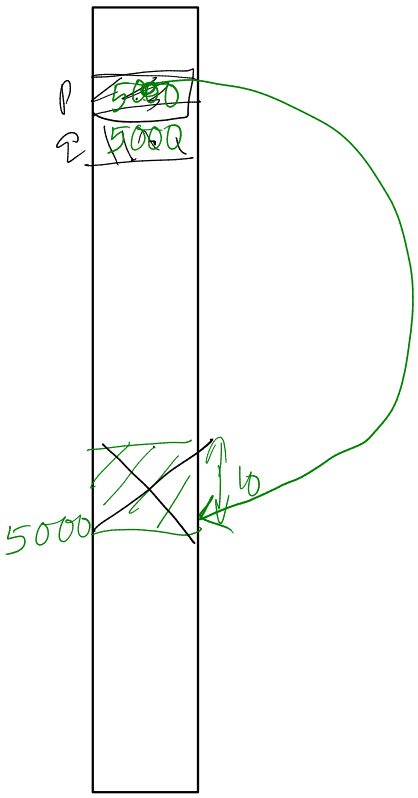
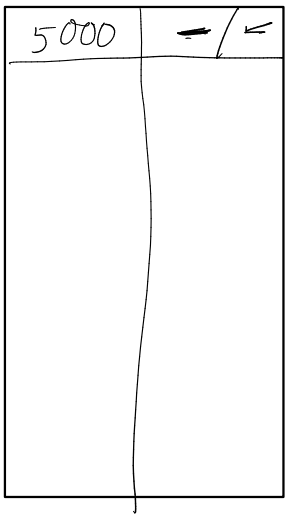
```
char *q;
char *p = NULL;
p = malloc(10);
q = p;
free(p);
```

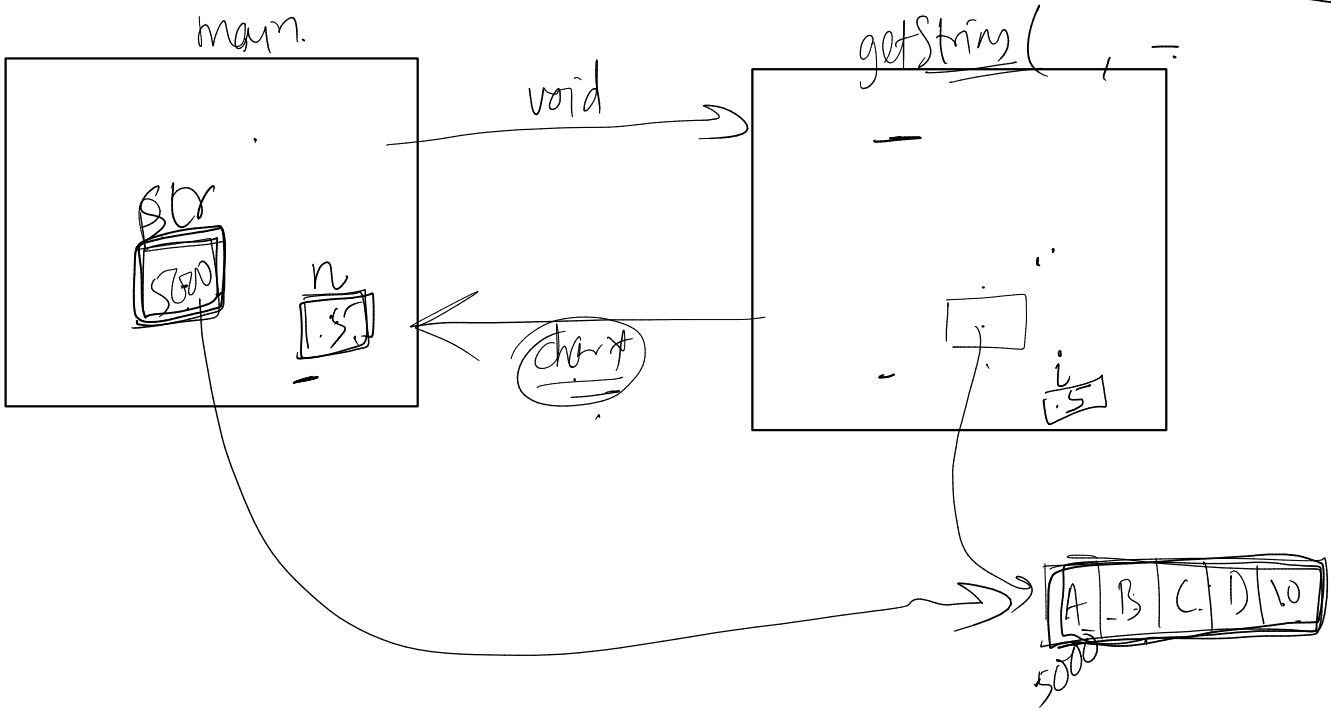


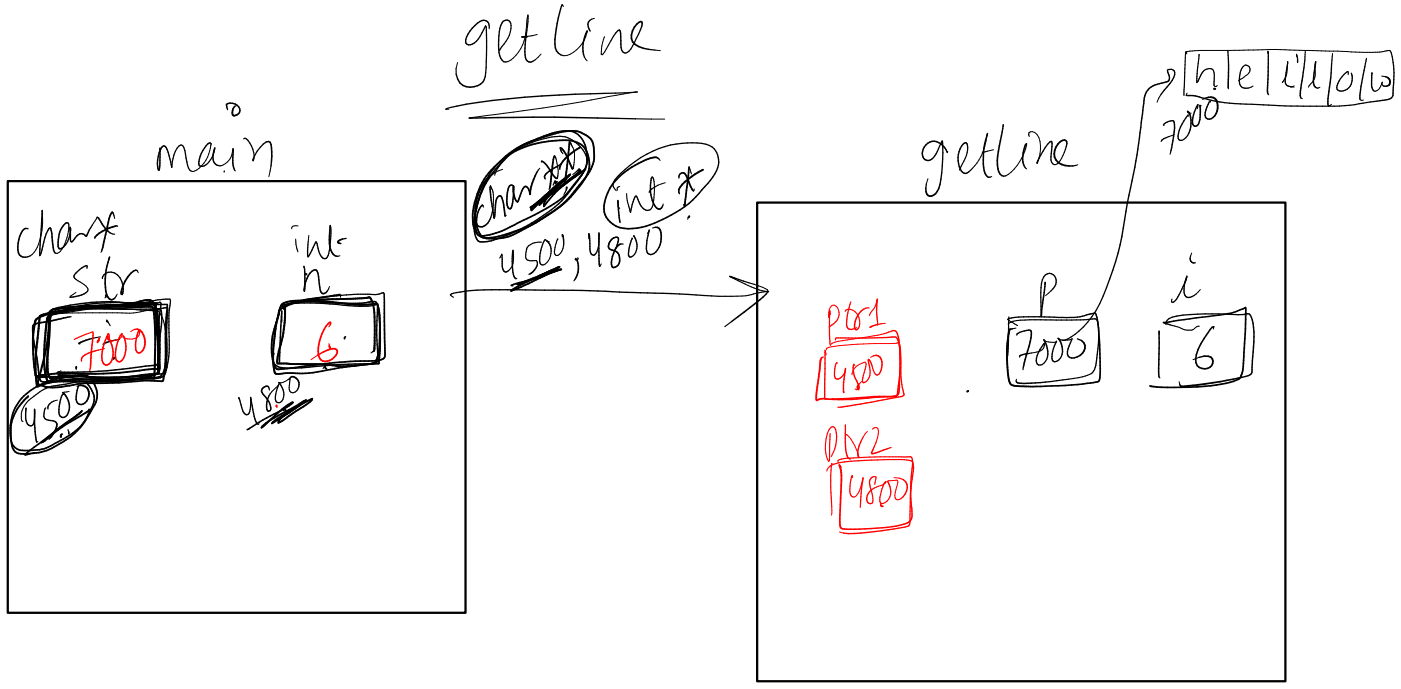


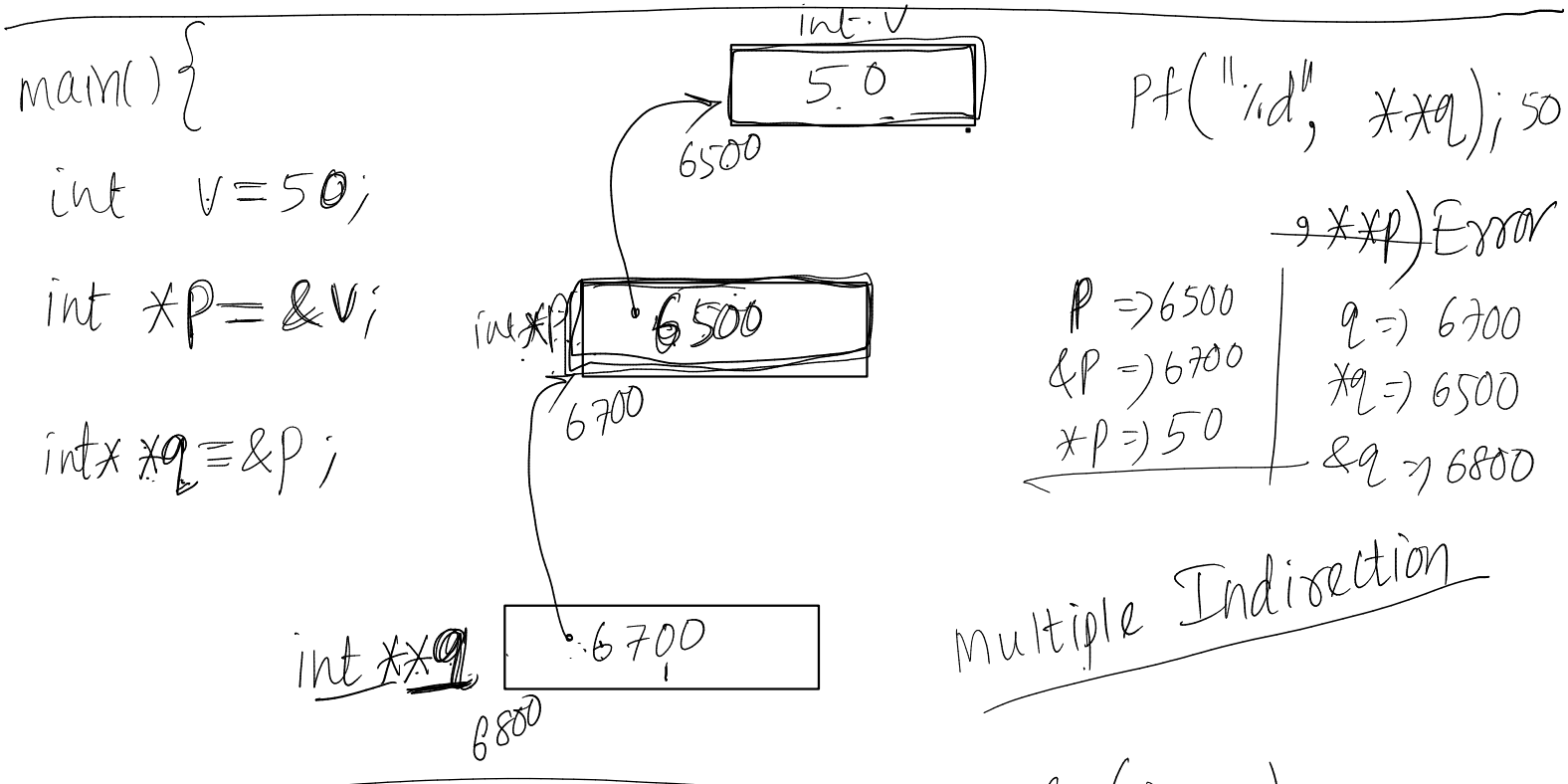

```
char *getstring(void);
```

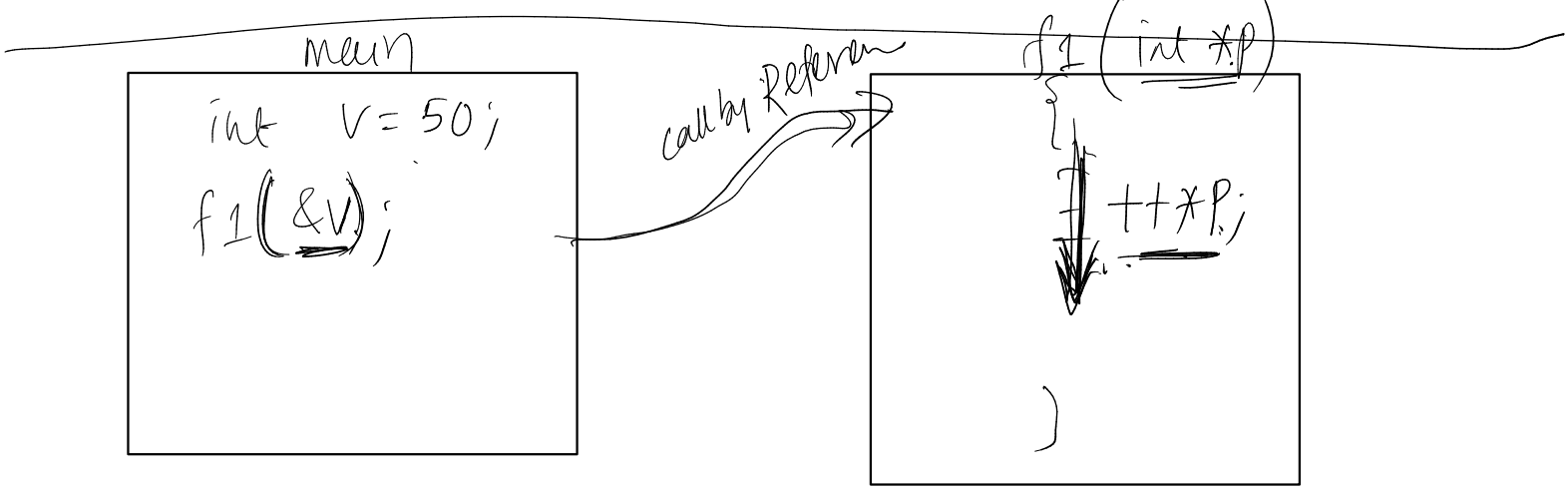
B8

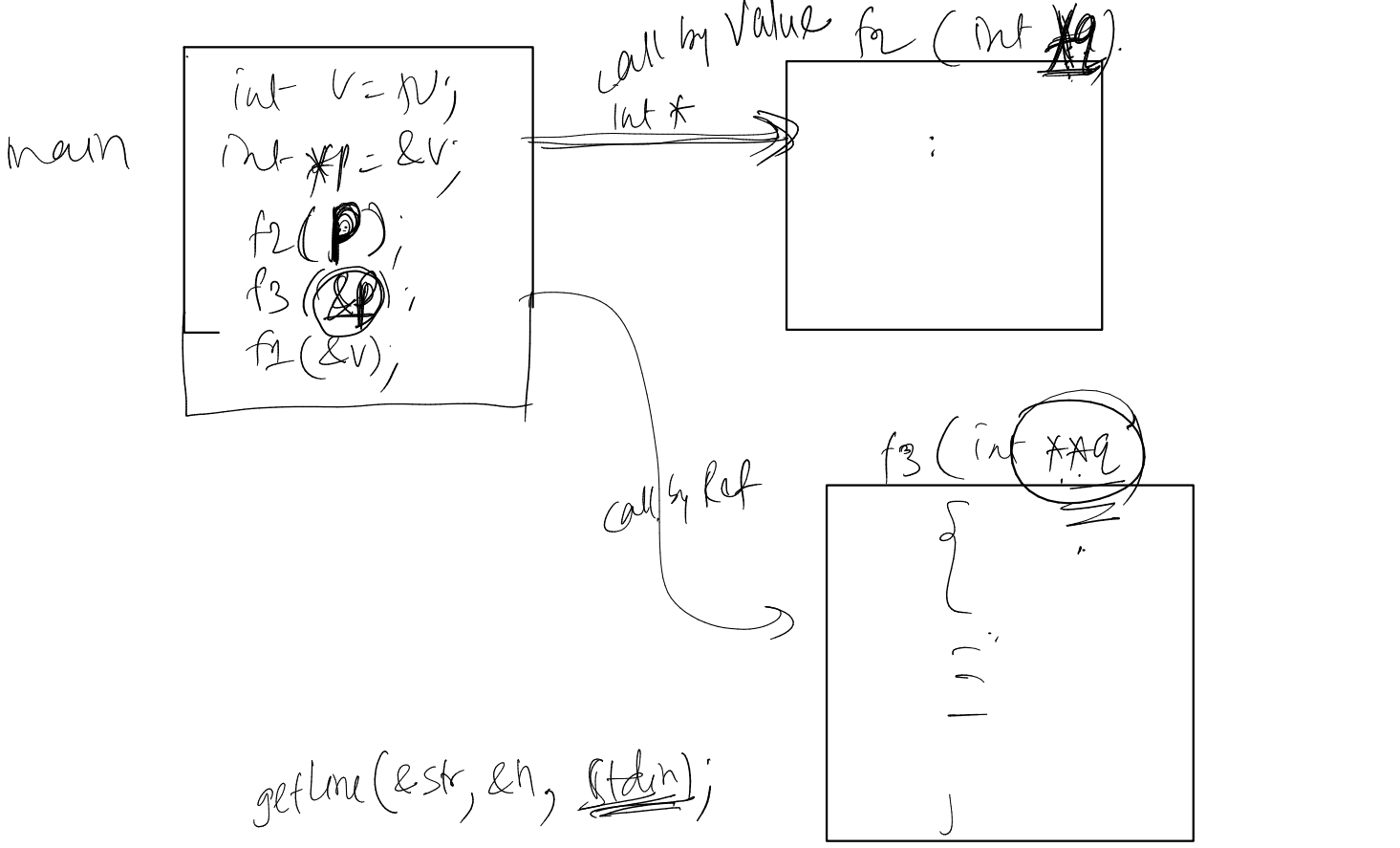
Description:

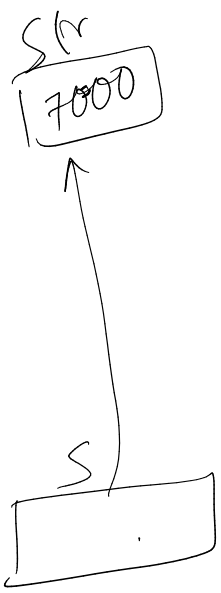



getline
main
char* str
7000
4500
int n
6
4800
char**
int *
4500, 4800
getline
h e l l o \0
7000
p
7000
i
6
ptr1
4500
ptr2
4800

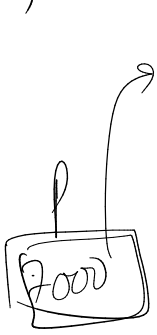

main(){
int V=50;
int *p=&V;
int **q=&p;
int V
50
6500
int *p
6500
6700
int **q
6700
6800
pf("%d", **q); 50
q **p) Error
P => 6500
&P => 6700
*P => 50
q => 6700
*q => 6500
&q => 6800
Multiple Indirection

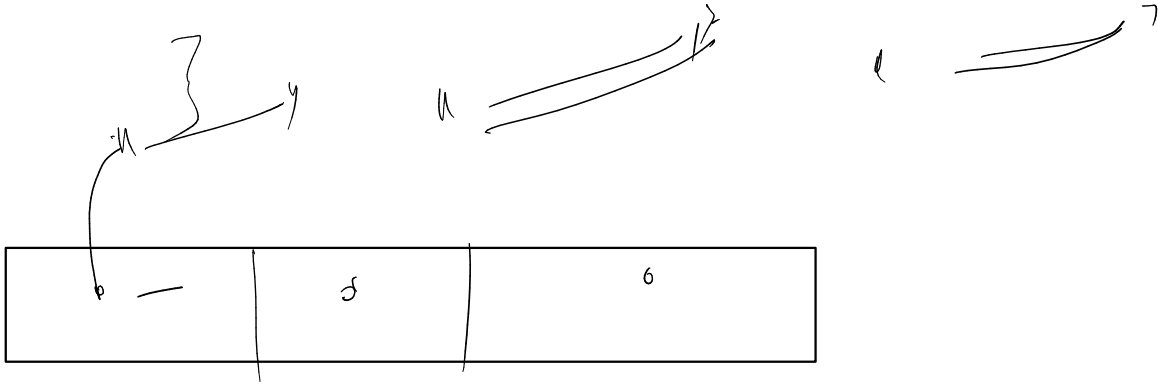

main
int V=50;
f1(&V);
call by Reference
f1(int *p)
++*p;
}






```
getline (char **s, int *n, FILE *fp)
{
char *p = NULL; int i = 0;
do {
   =
} while( - );
p[i] = '\0';
*s = p; *n = i;
```





2D Array

-> Array of Arrays

-> collection of multiple Arrays, in adjacent memory

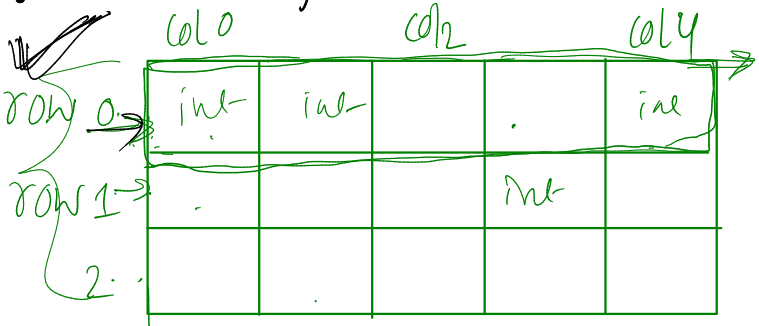
-> Benefit: A single Identifier, multiple Arrays.

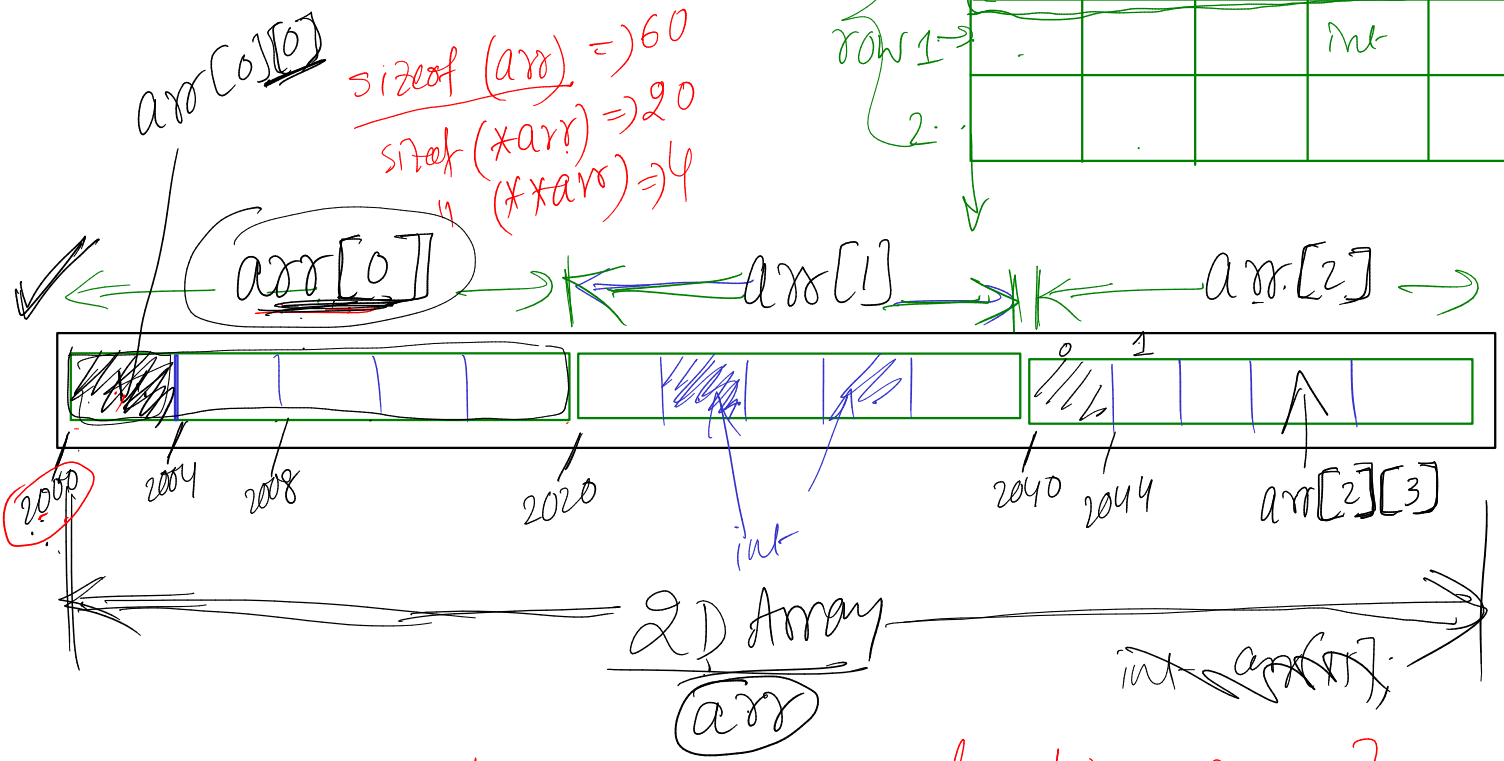
Syntax: datatype Arr_name [rowsize] [col-size];

e.g. int arr[3][5];

char names[10][20];

arr is a collection of 3 Arrays. Each Array is of 5 integers.





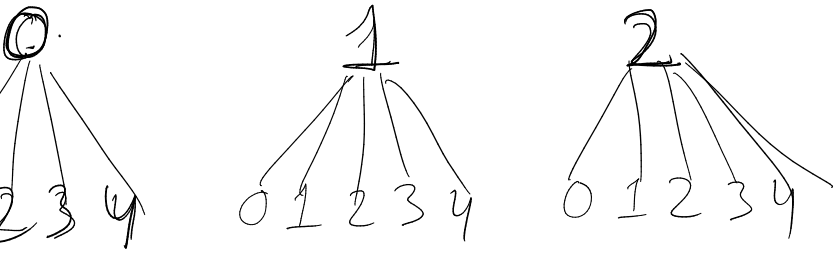
What type of data you see at location 2000?

-> 2D Array of - 60 bytes

-> 1D Array of - 20 bytes

-> integer of - 4 bytes

2D => i =>



1D => j => 0 1 2 3 4

rows = sizeof(arr) / sizeof(*arr); => 3

cols = sizeof(*arr) / sizeof(**arr); => 5

```
for(i=0; i<rows; i++, pf("\n"))
    for(j=0; j<cols; j++)
        pf("%d ", arr[i][j]);
```

W.A.P to store Marks in 5 Subjects, for 20 Students.

Find: 1) The student who scored Highest total

2) The subject, where maximum students failed.

NOTE: use rand() to collect the marks.

```
int st[20][5];
```

